Pouria Khakpour

Go Developer

Summary

A passionate Go developer specializing in efficient web applications using Gin and GORM, with strong network programming skills demonstrated by my implementation of a proxy protocol in Go. I excel in designing and implementing master-slave architectures for synchronization APIs across multiple servers, enhancing project maintainability through effective code refactoring and thorough documentation. As a self-taught developer, I quickly learn and apply new technologies pragmatically. My focus is on developing secure, scalable systems, with a keen interest in blockchain technology and its applications in secure digital finance. I am also eager to expand my knowledge of network programming and red-team cybersecurity.

Skills

Programming Languages

Go, Python, Rust, Bash

Frameworks/Libraries

Gin, GORM, Gorilla, Protocol Buffers, Flask, APIFlask

Databases

MySQL, SQLite, Redis

Tools

Git, Docker, Jinja2, Wireshark

Details

Location

Gilan, Rasht

Phone

+989338760704

Email

Pouria.Khakpour9909@gmail.com

GitHub

Github.com/0ne-zero

Education

Bagher-ol-Oloum High School

Masaal, Gilan May 2024

Languages

English

Professional working proficiency

Persian

Native

Hobbies

Music

Cafe-hopping

Experience

Python Developer, FOSS Project

Aug 2023 – Jun 2024 (11 months) – Remote

- Designed and implemented a **master-slave architecture** for syncing configuration and user data across multiple servers, ensuring reliable and consistent data synchronization.
- Developed **RESTful APIs** to facilitate communication between master and slave servers, incorporating strict validation rules for JSON input formats to ensure **data integrity** during interactions across multiple servers.
- Created a Python script using **Jinja2** to optimize the configuration generation process, eliminating configuration-related errors common with the previous Bash method and resulting in a **more reliable** and streamlined workflow.
- Refactored a codebase of over 5000 lines into modular Python packages, while providing **detailed documentation**, significantly enhancing code structure, readability, and facilitating **long-term maintainability**.
- Collaborated actively with testers, identifying and fixing bugs early, leading to **stable releases** and enhancing **production reliability.**

Freelance Projects

Go Programmer, GoTraceroute

Jul 2025

- Developed a cross-platform traceroute library and CLI tool in Go, supporting **IPv4/IPv6** and **TCP/UDP** protocols.
- Leveraged raw sockets and low-level **OS networking APIs** to craft custom **TCP SYN** and **UDP** probe packets, achieving control over outgoing traffic.
- Designed concurrent monitoring and accurate probe-response matching to enhance traceroute **speed** and **reliability**.
- Implemented **configurable** parameters including protocol, first TTL, max hops, IP version preference, timeout, retries, delay, and early stop conditions, making the tool **flexible**.

Go Programmer, Proxy Protocol

Sep 2024

- Designed and implemented a custom proxy protocol inspired by SOCKS5, handling encrypted traffic using AEAD ciphers for **end-to-end encryption**.
- Developed an **authentication mechanism** using AES/GCM encryption to securely validate the client-server handshake, effectively preventing **man-in-the-middle(MITM)** attacks, unauthorized access, and ensuring data integrity.
- Built a nonce-based system to protect against **replay attacks**, ensuring the integrity and confidentiality of communications within the protocol and enhancing the **anonymity**.
- Enhanced expertise in **Go programming**, encryption, **protocol development**, authentication systems and code documentation.

Translator, Easy Rust Fa

Feb 2023 – May 2023 (4 months)

- Translated comprehensive Rust language documentation with 8.1K stars on GitHub from English to Persian, demonstrating **technical translation skills**, deepening Rust language proficiency, and contributing to the programming community by providing valuable **localized resources**.

Rust Programmer, INID

Dec 2022

- Created a command-line program in Rust to **generate and validate** Iran's national identifier code (INID), gaining a strong understanding of Rust's syntax and features for developing command-line applications.

Go Programmer, Comic (Gin, GORM, MySQL)

Jun 2022

- Developed a comic website using **Gin and GORM**, incorporating features like commenting, liking, and searching to enhance user engagement while gaining proficiency in Go web development.

Go Programmer, Truth-Dare Bot

Jun 2022

- Developed a Telegram bot for the Truth-Dare game in Persian, enhancing my skills in **bot development** with Go.

Python Developer, Emma Text Assistant

Oct 2021 – Jan 2022

- Created a command-line text assistant featuring chat, Google search, text translation, Wikipedia search, date and time display, and joke telling capabilities.
- Utilized the **State design pattern** to manage mode transitions (Text, Half-Audio, CMD, and Stop), enhancing both functionality and code readability.

Courses

Bug Hunting, Voorivex Academy May 2025
Intro to Cybersecurity, Hunt Learn Sep. 2024
SEC542 Web App Penetration Testing, SANS Aug. 2024
Certified Ethical Hacker (CEH), Maktabkhooneh Jul. 2023
Machine Learning with Python, Maktabkhooneh Apr. 2023
Blockchain and Cryptocurrency(Bitcoin) Fundamentals, Maktabkhooneh Jun. 2022
Linux LPIC-2, Gotoclass Apr. 2022
Linux LPIC-1, Maktabkhooneh Feb. 2022